ProbOS — Month 3, Week 11
Pharmaceutical Stability Model

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Week 11 goal

Build `PharmaStabilityModel`, reusing `BatteryModel2Cell`'s Arrhenius reaction-kinetics architecture with drug potency as the depleting quantity, per the near-term priority established in `docs/monthly_plans/month3/main.tex` and the strategic reasoning in `docs/vision/main.tex`.

Critical constraint, stated up front: this model's credibility rests entirely on being validated against a REAL, genuine published stability-study dataset – the same standard Kim et al. (2007) set for `BatteryModel2Cell` in Month 1 Week 2. Synthetic or fabricated validation data would violate this project's established honesty discipline (`docs/standards/quality_standards.md`) and must not be used. If a suitable public dataset cannot be found by the end of Day 1, this week's scope will be explicitly narrowed (e.g. to model architecture and unit correctness only, with validation flagged as a genuinely open item) rather than proceeding with fabricated data.

Day-by-day plan

Day 1 – Literature research (before any model code)

- Search for a real, published pharmaceutical stability study reporting accelerated and/or long-term degradation data (ICH Q1A-style: potency vs. time at multiple temperatures), analogous to how Kim (2007)'s ARC data was used for the battery model
- Confirm the underlying degradation mechanism is genuinely Arrhenius (most small-molecule drug degradation is, but this must be confirmed for the SPECIFIC dataset chosen, not assumed)
- Document the citation precisely (matching the existing standard: exact source, exact reported values, no invented numbers) before writing any model code

Day 2 – PharmaStabilityModel implementation

- Design the state vector and parameter vector based on the ACTUAL structure of the dataset found on Day 1, not assumed in advance – unlike the battery model's 2-cell, 3-reaction structure, a real stability study may report a single degradation pathway, requiring a simpler state vector
- Implement `forward_batch()` following the exact same vectorised-NumPy, CPU/GPU dispatch-ready pattern established in `BatteryModel2Cell` (Week 10's `xp` dispatch and `cp.fuse()` pattern should be reused directly, not reinvented)

Day 3 – Validation against the real dataset

- Deterministic validation: run the model at the dataset's own reported nominal parameters, confirm it reproduces the reported potency-vs-time curve within a stated, justified tolerance – matching the standard already set for Kim (2007)
- If the model does NOT match the real data, this must be reported honestly and investigated, not hidden or the tolerance quietly loosened until it passes

Day 4 – Monte Carlo + Sobol sensitivity

- Build parameter priors reflecting genuine manufacturing/formulation uncertainty (informed by the dataset's own reported variability where available, not invented)
- Run `SobolSensitivity` to identify which parameters drive shelf-life uncertainty – the direct pharma analogue of `Ea_SEI`'s dominance in the battery model

Day 5 – Test coverage

- `python/tests/test_pharma_stability.py`: shape contracts, physical invariants (potency cannot exceed 100% or go negative), and the Day 3 validation-against-real-data test as the centerpiece, matching `test_battery_model.py`'s standard

Day 6 – pre_week_audit.sh + CI

- Run `pre_week_audit.sh` before considering the week complete
- Add the new model to CI's example-running steps

Day 7 – Documentation + retrospective

- Update `docs/vision/main.tex`: move pharmaceutical stability from "Structurally ready" to "Validated" IF Day 3's validation genuinely succeeds against real data – if it does not, leave the honest status as-is and document why
- Retrospective appended to this document once complete

Exit criteria

`PharmaStabilityModel` exists, is validated against a real, cited, published stability dataset (not fabricated data) at an honestly-justified tolerance, and its dominant uncertainty driver is identified via Sobol sensitivity – matching the exact validation discipline already established for `BatteryModel2Cell`. If a suitable real dataset cannot be found, this is reported honestly as a genuinely open item, not worked around with synthetic data presented as if it were real.

What was actually accomplished (retrospective)

- `PharmaStabilityModel` built and validated against a REAL, peer-reviewed, ICH-referenced dataset: Gonzalez-Gonzalez et al. (*Molecules* 2023, 28(23), 7925), chlorhexidine stability data.

Per the plan's own Day 1 gate, an initial candidate (aspirin hydrolysis kinetics from a 2024 teaching lab module) was explicitly rejected on closer scrutiny: real, peer-reviewed, and numerically correct, but pedagogical rather than an actual ICH Q1A-style product stability study.

- A genuine architectural finding emerged: the real, best-validated pharma dataset uses Avrami kinetics (a closed-form $potency = \exp(-kt^n)$ relationship), not the first-order depletion law used by `BatteryModel2Cell`. Code reuse happened at the architecture level (`Model` ABC, `xp` CPU/GPU dispatch, Sobol/provenance integration), not at the equation level.
- A genuine fitting failure was found and fixed: an unconstrained fit of both the Avrami pre-exponential factor (A) and exponent (n) against the paper's 3 real reported 365-day degradation percentages converged to a mathematically passing but physically absurd result ($A = 1.5 \times 10^{255}$, $n = -94.6$) – a textbook degenerate/underdetermined fit. Fixed by fixing $n = 1$ as a labeled simplifying assumption and fitting only A, which achieved an IDENTICAL residual error, confirming the extra degree of freedom was pure unidentifiable noise.
- The probabilistic core was reconnected to this new domain: using the paper's own reported Ea standard error (±2.61 kcal/mol, genuine literature-backed uncertainty), Sobol sensitivity identified Ea as overwhelmingly dominant ($S_1 = 0.993$), and Monte Carlo propagation revealed a real tail risk: the worst-case 5% of outcomes show complete potency loss within one year, versus 82.6% potency at the point estimate.

What went right

1. The Day 1 research gate worked as intended: scrutinizing the first candidate dataset caught a real quality gap before any model code was built around it.
2. A direct question about whether the probabilistic core still mattered led to genuinely reconnecting the deterministic groundwork to real Monte Carlo + Sobol analysis, rather than treating the curve-fit as the finish line.

What went wrong, and was fixed

1. An unconstrained 2-parameter fit produced a passing residual error alongside physically nonsensical parameter values. Caught by inspecting the actual fitted values, not just the error metric, and fixed by removing the underlying degeneracy.

Numbers at Week 11 close

- Python tests: 398/398 passing (12 new)
- mypy strict: 0 errors
- ruff: 0 warnings
- Coverage: 89.18%
- `pre_week_audit.sh`: 16/16 checks passed
- Validation accuracy: max 1.18 percentage points against 3 real reported data points
- Sobol dominant parameter: Ea ($S_1 = 0.993$)

Carried into future weeks

The Avrami exponent n remains a fixed simplifying assumption, not an independently recovered value. Automotive/EV safety positioning (also a Month 3 near-term priority) remains untouched – it requires no new model, only outreach positioning around the already-validated battery model.